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Batch : CH3

Subject : EDS

Screenshot Of Practical No.5 :

Practical Lab Assignment :

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5.1.1. Stacked Plot

Create a stacked area plot to visualize the temperature variations for three different cities (City A, City B, and City C) across the months of the year. The temperature data is provided for each city in the editor.

Your task is to:

- Create a stacked area plot using the data.
- Label the x-axis as "Month", the y-axis as "Temperature", and provide the title "Temperature Variation" for the plot.
- Display the plot showing the temperature variation for each city throughout the months of the year.

Sample Test Cases

stackedpl...

```
1 import matplotlib.pyplot as plt
2 import pandas as pd
3
4 # Data for Months and Temperature for three cities
5 data = {
6     'Month': ['January', 'February', 'March', 'April', 'May', 'June', 'July',
7             'August', 'September', 'October', 'November', 'December'],
8     'City_A_Temperature': [5, 7, 10, 13, 17, 20, 22, 21, 18, 12, 8, 6],
9     'City_B_Temperature': [2, 3, 5, 6, 10, 14, 16, 17, 12, 9, 5, 3],
10    'City_C_Temperature': [3, 4, 6, 8, 9, 12, 15, 14, 10, 7, 4, 2]
11 }
12
13 # Write your code...
14 df = pd.DataFrame(data)
15
16 plt.stackplot(df['Month'], df['City_A_Temperature'], df['City_B_Temperature'],
17              df['City_C_Temperature'])
```

Average time: 0.155 s, Maximum time: 155.00 ms, 1 out of 1 shown test case(s) passed

Test case 1: 155 ms, Debug, Test cases

Expected output:

Actual output:

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5.2.1. Titanic Dataset

Write a Python program to analyze and visualize data from the Titanic dataset based on the following instructions:

Dataset Information:

The dataset is stored in a CSV file named `titanic.csv` and has been loaded using the pandas library. It contains the following columns:

- Pclass: Passenger class (1 = First, 2 = Second, 3 = Third).
- Gender: Gender of the passenger (male/female).
- Age: Age of the passenger.
- Survived: Survival status (0 = Did not survive, 1 = Survived).
- Fare: Ticket fare paid by the passenger.

Visualization:

To represent these trends, you will create 5 visualizations using Matplotlib. The visualizations should be arranged in a 3x2 grid (3 rows and 2 columns).

Visualization Details:

Write the code to create a series of visualizations as follows:

Bar Plot (Pclass Distribution):

- Create a bar plot to show the distribution of passengers across the different passenger classes (Pclass).
- Use the color skyblue for the bars.
- Title the plot as "Passenger Class Distribution".
- Label the x-axis as "Pclass" and the y-axis as "Count".

Pie Chart (Gender Distribution):

- Create a pie chart to display the distribution of male and female passengers.
- Use lightblue for males and lightcoral for females.
- Include percentages on the slices (use `autopct='%1.1f%%'`).
- Title the plot as "Gender Distribution".

Histogram (Age Distribution):

titanicDat...

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset from the csv file
5 df = pd.read_csv('titanic.csv')
6
7 # Set up the figure for 5 subplots
8 fig, axes = plt.subplots(3, 2, figsize=(12, 12))
9
10 # write the code..
11 axes[0,0].bar(df['Pclass'].value_counts().index, df['Pclass'].value_counts(), color='skyblue')
12 axes[0,0].set_title("Passenger Class Distribution")
13 axes[0,0].set_xlabel("Pclass")
14 axes[0,0].set_ylabel("Count")
15
16 axes[1,0].pie(df['Gender'].value_counts().index, df['Gender'].value_counts().index,
17              autopct='%1.1f%%', colors=['lightblue', 'lightcoral'])
18 axes[1,0].set_title("Gender Distribution")
19 axes[1,0].set_xlabel("Gender")
20 axes[1,0].set_ylabel("Count")
```

Average time: 0.586 s, Maximum time: 586.00 ms, 1 out of 1 shown test case(s) passed

Test case 1: 586 ms, Debug, Test cases

Expected output:

Actual output:

8.2.2. Histogram of passenger information of Titanic

Write a Python code to plot a histogram for the distribution of the 'Age' column from the Titanic dataset. The histogram should display the frequency of different age ranges with the following specifications:

- Use **30 bins** for the histogram.
- Set the **edge color** of the bars to **black (k)**.
- Label the x-axis as **'Age'** and the y-axis as **'Frequency'**.
- Add the title **"Age Distribution"** to the histogram.

The Titanic dataset contains columns as shown below,

Passenger Id	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```

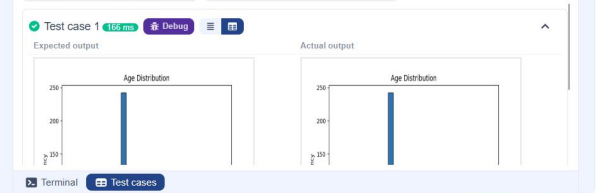
PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,0,0,338877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achen)",female,14,1,0,237736,30.0708,,C
    
```

Note: Refer to the visible test case for better reference.

Histogram...

```

1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 # Write your code here for Histogram
17 plt.hist(data['Age'], bins=30, edgecolor='k')
    
```



8.2.3. Bar plot of survival rate of passengers

Write a Python code to plot a bar chart that shows the count of passengers who survived and did not survive in the Titanic dataset. The chart should display the following specifications:

- Use the **'Survived'** column to show the count of survivors (0 = Did not survive, 1 = Survived).
- Set the chart type to **'bar'**.
- Add the title **"Survival Count"** to the chart.
- Label the x-axis as **'Survived'** and the y-axis as **'Count'**.

The Titanic dataset contains columns as shown below,

Passenger Id	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```

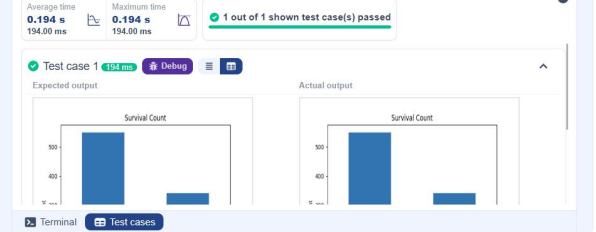
PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,0,0,338877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achen)",female,14,1,0,237736,30.0708,,C
    
```

Note: Refer to the visible test case for better reference.

BarPlotOf...

```

1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 # Write your code here for Bar Plot for Survival Rate
17 plt.bar(data['Survived'], data['Survived'].count(), title='Survival Count')
    
```



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6.2.4. Bar Plot for Survival by Gender

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Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by gender, in the Titanic dataset. The chart should display the following specifications:

1. Group the data by the **Sex** column, then use the **value_counts()** function to count the occurrences of survivors (0 = Did not survive, 1 = Survived) for each gender.

2. Use a **stacked bar chart** to display the survival counts.

3. Add the title **"Survival by Gender"** to the chart.

4. Label the x-axis as **Gender** and the y-axis as **Count**.

5. The legend should indicate **'Not Survived'** and **'Survived'**.

The Titanic dataset contains columns as shown below,

Passenger Id	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked

1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,S

2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C

3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2 3101282,7.925,S

4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S

5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,S

6,0,3,"Moran, Mr. James",male,0,0,330877,0.4533,Q

7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S

8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349900,21.075,S

9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,S

10,1,2,"Nasser, Mrs. Nicholas (Adele Achen)",female,14,1,0,237736,30.0706,C

Note: Refer to the visible test case for better reference.

BarPlotOf...

1import pandas as pd

2import matplotlib.pyplot as plt

3

4# Load the Titanic dataset

5data = pd.read_csv("Titanic-Dataset.csv")

6

7# Data Cleaning

8data["Age"].fillna(data["Age"].median(), inplace=True)

9data["Embarked"].fillna(data["Embarked"].mode()[0], inplace=True)

10data.drop("Cabin", axis=1, inplace=True)

11

12# Convert categorical features to numeric

13data["Sex"] = data["Sex"].map({'male': 0, 'female': 1})

14data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

15

16# Write your code here for Bar-Plot for Survival by Gender

17survival_counts = data.groupby('Sex')['Survived'].value_counts().unstack()

Average time0.210 s

Maximum time0.210 s

210.00 ms

210.00 ms

1 out of 1 shown test case(s) passed

Test case 1210 ms

Debug

Expected output

Actual output

Survival by Gender

Survival by Gender

Not Survived

Survived

Not Survived

Survived

Terminat

Test cases

< Prev

Reset

Submit

Next >

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6.2.6. Bar Plot for Survival by Pclass

Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by passenger class (Pclass), in the Titanic dataset. The chart should display the following specifications:

- Group the data by the Pclass column and count the number of survivors (0 = Did not survive, 1 = Survived) for each class using value_counts().
- Use a stacked bar chart to display the survival counts.
- Add the title "Survival by Pclass" to the chart.
- Label the x-axis as "Pclass" and the y-axis as "Count".
- The legend should indicate "Not Survived" and "Survived".

The Titanic dataset contains columns as shown below,

Passenger Id	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked											
1,0,3,"Braund, Mr. Owen Harris"	male,22,1,0,A/5	21171,7.25,,5	2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)"	female,38,1,0,PC	17599,71.2833,C85,C	3,1,3,"Heikkinen, Miss. Laina"	female,26,0,0,STON/O2	3101282,7.925,,5	4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)"	female,35,1,0,113803,53.1,C123,S	5,0,3,"Allen, Mr. William Henry"	male,35,0,0,373458,8.05,,5	6,0,3,"Moran, Mr. James"	male,,0,0,330877,0.4583,,Q	7,0,1,"McCarthy, Mr. Timothy J"	male,54,0,0,17463,51.8625,E46,S	8,0,3,"Palsson, Master. Gosta Leonard"	male,2,3,1,349090,21.075,,S	9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)"	female,27,0,2,347742,11.1333,,S	10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)"	female,14,1,0,237736,30.8708,,C

Sample Data:

```
PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,5
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2 3101282,7.925,,5
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373458,8.05,,5
6,0,3,"Moran, Mr. James",male,,0,0,330877,0.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349090,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.8708,,C
```

Note:

BarPlotOf...

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 # Write your code here for Bar Plot for Survival by Pclass
17 survival_count = data.groupby('Pclass').
```

Average time0.217 sMaximum time0.217 s1 out of 1 shown test case(s) passed

Test case 10.217 msDebug

Expected outputActual output

Survival by PclassSurvival by Pclass

TerminalTest cases

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6.2.6. Bar Plot for Survival by Embarked

Write a Python code to plot a stacked bar chart showing the survival count for passengers based on their embarkation location in the Titanic dataset. The chart should display the following specifications:

- Use the Embarked column to determine the embarkation location. After converting this column into dummy variables (using pd.get_dummies()), plot the survival count based on the Embarked_Q column (representing passengers who embarked from Queenstown) in relation to survival.
- Set the chart type to 'bar' and make it stacked.
- Add the title "Survival by Embarked" to the chart.
- Label the x-axis as "Embarked" and the y-axis as "Count".
- Include a legend to distinguish between survivors and non-survivors (label the legend as "Survived" and "Not Survived").

The Titanic dataset contains columns as shown below,

Passenger Id	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked											
1,0,3,"Braund, Mr. Owen Harris"	male,22,1,0,A/5	21171,7.25,,5	2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)"	female,38,1,0,PC	17599,71.2833,C85,C	3,1,3,"Heikkinen, Miss. Laina"	female,26,0,0,STON/O2	3101282,7.925,,5	4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)"	female,35,1,0,113803,53.1,C123,S	5,0,3,"Allen, Mr. William Henry"	male,35,0,0,373458,8.05,,5	6,0,3,"Moran, Mr. James"	male,,0,0,330877,0.4583,,Q	7,0,1,"McCarthy, Mr. Timothy J"	male,54,0,0,17463,51.8625,E46,S	8,0,3,"Palsson, Master. Gosta Leonard"	male,2,3,1,349090,21.075,,S	9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)"	female,27,0,2,347742,11.1333,,S	10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)"	female,14,1,0,237736,30.8708,,C

Sample Data:

```
PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,5
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2 3101282,7.925,,5
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373458,8.05,,5
6,0,3,"Moran, Mr. James",male,,0,0,330877,0.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349090,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.8708,,C
```

Note:

BarPlotOf...

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 # Write your code here for Bar Plot for Survival by Embarked
17 data.head()
```

Average time0.232 sMaximum time0.232 s1 out of 1 shown test case(s) passed

Test case 10.232 msDebug

Expected outputActual output

Survival by EmbarkedSurvival by Embarked

TerminalTest cases

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6.2.7. Box plot for Age Distribution

Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset across different passenger classes. The boxplot should display the following specifications:

- Use the Pclass column to group the data for the boxplot.
- Set the title of the plot to "Age by Pclass".
- Remove the default subtitle with plt.suptitle('').
- Label the x-axis as "Pclass" and the y-axis as "Age".

The Titanic dataset contains columns as shown below,

Passenger Id	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked											
1,0,3,"Braund, Mr. Owen Harris"	male,22,1,0,A/5	21171,7.25,,5	2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)"	female,38,1,0,PC	17599,71.2833,C85,C	3,1,3,"Heikkinen, Miss. Laina"	female,26,0,0,STON/O2	3101282,7.925,,5	4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)"	female,35,1,0,113803,53.1,C123,S	5,0,3,"Allen, Mr. William Henry"	male,35,0,0,373458,8.05,,5	6,0,3,"Moran, Mr. James"	male,,0,0,330877,0.4583,,Q	7,0,1,"McCarthy, Mr. Timothy J"	male,54,0,0,17463,51.8625,E46,S	8,0,3,"Palsson, Master. Gosta Leonard"	male,2,3,1,349090,21.075,,S	9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)"	female,27,0,2,347742,11.1333,,S	10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)"	female,14,1,0,237736,30.8708,,C

Sample Data:

```
PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,5
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2 3101282,7.925,,5
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373458,8.05,,5
6,0,3,"Moran, Mr. James",male,,0,0,330877,0.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349090,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.8708,,C
```

Note: Refer to the visible test case for better reference.

BoxPlotFor...

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 # Write your code here for Box Plot for Age by Pclass
17 data.head()
```

Average time0.214 sMaximum time0.214 s1 out of 1 shown test case(s) passed

Test case 10.214 msDebug

Expected outputActual output

Age by PclassAge by Pclass

TerminalTest cases

CODETANTRAHomeLearn Anywhere202501050062@mitaoe.ac.inSupportLogout

5.2.8. Box Plot for Age by Survived

Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset based on whether passengers survived or not. The boxplot should display the following specifications:
1. Use the **Survived** column to group the data for the boxplot (0 = Did not survive, 1 = Survived).
2. Set the title of the plot to **"Age by Survival"**.
3. Remove the default subtitle with **plt.suptitle("")**.
4. Label the x-axis as **'Survived'** and the y-axis as **'Age'**.

The Titanic dataset contains columns as shown below,

Passenger Id	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2 3101282,7.925,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,S
6,0,3,"Moran, Mr. James",male,0,0,330877,8.4583,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349009,21.075,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achen)",female,14,1,0,237736,30.0786,C

Note: Refer to the visible test case for better reference.

BoxPlotF...

1import pandas as pd
2import matplotlib.pyplot as plt
3
4# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')
5
6# Data Cleaning
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)
7
8# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
9
10# Write your code here for Box Plot for Age by Survived
plt.figure(figsize=(8,6))
11
12
13
14
15
16
17

Average time0.195 sMaximum time0.195 s1 out of 1 shown test case(s) passed

Test case 1195 msDebug

Expected outputActual output

Age by Survival

TerminalTest cases

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5.2.9. Box Plot for Fare by Pclass

Write a Python code to plot a boxplot that shows the distribution of the 'Fare' column from the Titanic dataset based on the passenger class (Pclass). The boxplot should display the following specifications:
1. Use the **Pclass** column to group the data for the boxplot.
2. Set the title of the plot to **"Fare by Pclass"**.
3. Remove the default subtitle with **plt.suptitle("")**.
4. Label the x-axis as **'Pclass'** and the y-axis as **'Fare'**.

The Titanic dataset contains columns as shown below,

Passenger Id	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2 3101282,7.925,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,S
6,0,3,"Moran, Mr. James",male,0,0,330877,8.4583,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349009,21.075,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achen)",female,14,1,0,237736,30.0786,C

Note: Refer to the visible test case for better reference.

BoxPlotF...

1import pandas as pd
2import matplotlib.pyplot as plt
3
4# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')
5
6# Data Cleaning
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)
7
8# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
9
10# Write your code here for Box Plot for Fare by Pclass
plt.figure(figsize=(8,6))
11
12
13
14
15
16
17

Average time0.180 sMaximum time0.180 s1 out of 1 shown test case(s) passed

Test case 1180 msDebug

Expected outputActual output

Fare by Pclass

TerminalTest cases

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5.2.10. Scatter Plot for Age vs. Fare

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset. The scatter plot should display the following specifications:
1. Use the **Age** column for the x-axis and the **Fare** column for the y-axis.
2. Set the title of the plot to **"Age vs. Fare"**.
3. Label the x-axis as **'Age'** and the y-axis as **'Fare'**.

The Titanic dataset contains columns as shown below,

Passenger Id	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2 3101282,7.925,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,S
6,0,3,"Moran, Mr. James",male,0,0,330877,8.4583,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349009,21.075,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achen)",female,14,1,0,237736,30.0786,C

Note: Refer to the visible test case for better reference.

Sample Test Cases

AgeFareS...

1import pandas as pd
2import matplotlib.pyplot as plt
3
4# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')
5
6# Data Cleaning
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)
7
8# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
9
10# Write your code here for Box Plot for Fare by Pclass
plt.scatter(data['Age'], data['Fare'])
11
12
13
14
15
16
17

Average time0.158 sMaximum time0.158 s1 out of 1 shown test case(s) passed

Test case 1158 msDebug

Expected outputActual output

Age vs. Fare

TerminalTest cases

CODETANTRAHomeLearn Anywhere202501050062@mitaoe.ac.inSupportLogout

5.2.11. Scatter Plot for Age vs. Fare by Survived

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset, with points color-coded by survival status. The scatter plot should display the following specifications:
1. Use the **Age** column for the x-axis and the **Fare** column for the y-axis.
2. Color the points based on the **Survived** column: **Red** for passengers who did not survive (**Survived = 0**). **Blue** for passengers who survived (**Survived = 1**).
3. Set the title of the plot to **"Age vs. Fare by Survival"**.
4. Label the x-axis as **'Age'** and the y-axis as **'Fare'**.

The Titanic dataset contains columns as shown below.

Passenger Id	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,3,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.65,,S
6,0,3,"Moran, Mr. James",male,0,0,338877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349989,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achen)",female,14,1,0,237736,30.0706,,C

Note: Refer to the visible test case for better reference.

AgeFareS...

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 colors = data['Survived'].map({0: 'red', 1: 'blue'})
17 plt.scatter(data['Age'], data['Fare'], c=colors)
```

Average time0.171 sMaximum time0.171 s1 out of 1 shown test case(s) passed

Test case 14171 msDebugTest cases

Expected outputActual output

Age vs. Fare by SurvivalAge vs. Fare by Survival

TerminatTest cases

PrevResetSubmit